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A Bayesian Deep Neural Network (DBNN) for Music Genre Classification

A multimodal CNN+ Bayesian neural network for music genre classification, demonstrating that reliability-aware metrics integrating accuracy, uncertainty, and stability provide a more principled basis for model selection than accuracy alone.

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Note(s):

A figure designated with no encoded lettering (e.g. Table 1) – appears in the main body of the text.
A figure encoded with lettering (e.g. Figure B1) – appears in appendix B.

Introduction

Automatic music genre classification is a key challenge in music information retrieval, demanding models that capture rhythmic, harmonic, and timbral structures across time. Using the GTZAN dataset—1,000 thirty-second audio clips across 10 genres such as classical, jazz, pop, and rock—each track is represented as a spectrogram that visualizes sound energy over time and serves as the input to a deep neural network.

This project, **B.E.A.R. D.O.W.N. — Bayesian Evaluation of Audio Representations: A Deep Optimized Waveform Network**, aims to develop a Bayesian Deep Neural Network (BDNN) that achieves 75-80% genre-recognition accuracy while providing calibrated uncertainty estimates. The study addresses **three goals**:

1. Evaluate whether a hybrid Bayesian / non-Bayesian model trained on 30-second GTZAN clips can reach **75–80% accuracy** using **k=3 cross-validation** and a broad **random hyperparameter sweep**, while recording full performance metrics (precision, recall, F1, per-genre metrics, ROC curves, confusion matrices, and Bayesian uncertainty). Apply Bayesian inference to the network’s learned embeddings to explain which spectral features drive genre distinctions and confusions.
2. After analyzing Sweep 1, refine and narrow the hyperparameter ranges and perform a second targeted sweep to determine whether accuracy can be **increased by an additional 2–3%**, using the same k-fold framework and expanded metric set (including RRM – a new metric for global accuracy).
3. Along with goal 2 test and define a new metric **RRM – Robust Ranking Metric**.

The **Robust Ranking Metric** (RRM) (Figure B1) is motivated by converging insights from literature.

- Bayesian neural networks naturally estimate predictive uncertainty, meaning model comparison should reflect both accuracy and epistemic variability rather than accuracy alone [3], [4].
- Cross-validation research shows that mean fold accuracy is insufficient on its own; variation across folds provides essential information about model stability and reliability [2], [5].
- Modern model-assessment frameworks emphasize jointly evaluating fit and uncertainty, especially when multiple candidate models achieve similar accuracy levels [1], [6].
- RRM combines these principles by integrating mean accuracy, fold-wise stability, and Bayesian predictive uncertainty into a single ranking measure that favors both strong performance and consistent reliability

Together, these goals establish a two-stage training and evaluation framework that uses broad and refined sweeps, full genre-level diagnostics, and tests a new RRM metric to identify models that not only achieve high accuracy but also demonstrate stable cross-fold performance and well-calibrated Bayesian uncertainty.

Code

Code, a readme, project description and usage statements are located here:

https://github.com/N-Herling-Mk1/INFO_510_Fa25_Final_Proj

Methods – for both questions

The **GTZAN Genre Collection** served as the foundation for this study, providing 1,000 audio tracks evenly (while the tracks were initially evenly distributed, some asymmetries occurred due to feature construction) distributed across ten musical genres (blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, rock). Although the dataset includes both 30-second and 10-second clips with corresponding feature tables, **only the 30-second clips were used** in this project – leaving the door open for further development with the 10 sec. clips. From each clip, two complementary data modalities were derived: a **spectrogram image** and a **set of 58 engineered audio features** (Appendix C, Table 3C). Spectrograms were generated using a Short-Time Fourier Transform (STFT) and standardized to a fixed resolution for convolutional processing, while tabular descriptors capturing chroma statistics, MFCC aggregates, spectral energy measures, and temporal dynamics were extracted in parallel.

Not all audio files successfully yielded both modalities. Some spectrograms failed due to input corruption or STFT inconsistencies, and some feature vectors contained incomplete values. After merging modalities, approximately **10% of samples could not be aligned** and were excluded, leaving roughly **900 fully matched multimodal samples** for model development.

Each modality underwent domain-appropriate preprocessing. Spectrograms were resized, normalized by pixel intensity, and optionally clipped to reduce extreme-frequency artifacts. Light augmentation—random time and frequency masking and small amplitude jitter—was applied during training to promote generalization without distorting genre-defining structure. Tabular features required no imputation, as samples with missing values were removed during alignment; instead, features were standardized via z-score normalization, and extreme outliers in spectral flux and energy-related measures were mitigated through winsorization. After preprocessing, spectrogram tensors and normalized 58-dimensional feature Table. 3C vectors formed the two independent model inputs.

The multimodal architecture comprised **three components**: Dual Encoders, a Fusion Block, and a Bayesian Classification Head (Figure 1; Appendix C, Table 1C). The Dual Encoders consisted of two deterministic subnetworks. The spectrogram pathway employed a compact convolutional architecture with convolution–ReLU–pooling blocks followed by flattening and a dense projection, while the tabular pathway used a small multilayer perceptron with ReLU activation and structured dropout to project the 58 features into a complementary latent embedding. These representations were forwarded to the Fusion Block. Fusion was performed using either simple concatenation or a gated mechanism, depending on the sampled hyperparameter configuration. This block remained deterministic, producing a unified multimodal representation that served as input to the probabilistic final layer.

The **Bayesian Classification Head** produced class probabilities along with estimates of epistemic and aleatoric uncertainty. Two Bayesian variants were supported: a **Flipout Dense layer** implementing variational inference and an **MC-Dropout Dense layer** approximating uncertainty via repeated stochastic forward passes. Both yielded softmax probabilities for the ten genres and uncertainty estimates aggregated during evaluation. The mk5m.py (sweep 1) training script extended mk5l.py (sweep 2) by computing additional diagnostics—including the **Residual Risk Metric (RRM)**, off-diagonal noise, and Monte Carlo stability indicators—without modifying the underlying architecture.

Training was conducted using randomized hyperparameter sweeps (Tables A1 and B2). Monte Carlo (Monte Carlo sampling via stochastic forward passes vs. full Markov Chain Monte Carlo) evaluation was performed post-training to estimate predictive entropy, per-genre uncertainty, and fold-stability measures. Only samples containing both modalities were evaluated; thus, the 10% data loss reduced sample size but avoided missing-feature bias. Inspection confirmed no disproportionate genre loss, mitigating concerns of genre-specific sampling bias.

Dataset – Selection and provenance

The dataset can be found at:

<https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification>

The original dataset consists of:

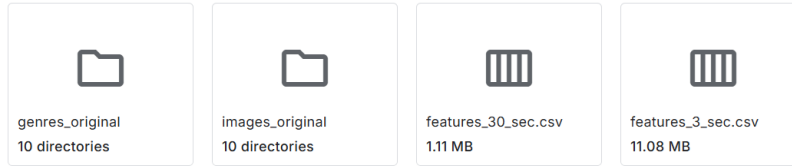


Figure 1. Topology of the GTZAN dataset.

The dataset is composed of audio directories and accompanying feature files: *genres_original* (10 genre folders of 30-second audio clips), *images_original* (10 directories of color spectrograms), and two CSV files containing pre-computed audio features for the 30-second and 3-second segments.

Note — the *images_original* directory contains color spectrogram representations of each track.

EDA plan

EDA will consist of checking for – (1) complete data set, (2) complete feature set. Note data set was ~99% complete, entries with empty features that could not be used in the training were simply skipped over.

When it came to splice the spectrogram files together with the features in the .csv, different components of the data, missing as either a spectrogram or a .csv feature ended up causing a ~10% gap in the total data. An extra function was constructed to ensure the validation set for all runs would have at least 90% coverage.

Question – 1 – Exploration and Analysis:

1.a - Introduction

Question 1 evaluates whether a **hybrid Bayesian / non-Bayesian multimodal neural network**, trained on **30-second clips from the GTZAN Genre Collection**, can achieve **75–80% classification accuracy** under **k = 3 cross-validation** using a **broad randomized hyperparameter sweep**. Each model ingests two complementary representations derived from the dataset: a spectrogram image and a corresponding set of engineered audio features. Only samples for which both modalities are available are used in training and evaluation. Question 1 explores accuracy possibilities under hyperparameter sweeps, and records Bayesian model metrics in order to explore their predictive potential in model validation.

For each candidate configuration, we compute a comprehensive set of performance metrics—including overall accuracy, precision, recall, F1, per-genre ROC curves, confusion matrices, and Bayesian posterior summaries of classifier-head parameters—to determine whether the target accuracy band can be reached and whether the Bayesian components behave sensibly during training. At this stage, Bayesian outputs are treated strictly as diagnostic signals rather than interpretive tools: epistemic uncertainty, genre-confusion structure, and reliability-aware ranking are intentionally deferred to Question 2. This question is of interest because it establishes a performance baseline for the hybrid architecture and tests whether conventional accuracy metrics alone are sufficient for assessing model quality in a Bayesian setting.

1.b Methods

To address **Question 1**, the model outlined in the ‘Methods – for both questions’ section was applied.

Models were trained and evaluated using **3-fold cross-validation** with a **broad randomized hyperparameter sweep (n = 20)** to explore architectural, optimization, and Bayesian design choices. Sampled parameters/hyperparameters – both frequentist and Bayesian are described in **Appendix A, Table A1**.

For Bayesian configurations, **priors were applied only to the classification head**. Flipout layers employed zero-mean Gaussian priors with fixed variance, while MC-Dropout approximated Bayesian inference via

stochastic forward passes. These priors regularize final-layer weights without constraining deterministic encoders; full prior specifications are provided in **Appendix C, Table 2C**.

Because inference relies on **variational and Monte Carlo approximations rather than full MCMC**, chain-based diagnostics (e.g., trace plots, Gelman–Rubin statistics) are not applicable. Instead, convergence and stability were assessed using training-curve behavior, fold-to-fold consistency, and repeated Monte Carlo sampling. Posterior summaries and **per-class credible intervals** were derived from stochastic forward passes, with interval definitions given in **Appendix A, Table A2**.

Predictive performance was evaluated using **standard classification metrics** (accuracy, per-genre precision, recall, F1, macro/micro averages, ROC curves, and confusion matrices). To address the ~10% sample loss during spectrogram–feature alignment (see Main Methods), a **holdout-coverage constraint** ensured adequate genre representation within each fold. At this stage, Bayesian uncertainty was recorded **only as a diagnostic signal**, not as a ranking or interpretive metric.

Overall, **Question 1 is exploratory**, assessing whether the hybrid architecture can reach the **75–80% accuracy target** under conventional validation and identifying failure modes when accuracy-centric metrics are used in isolation. Uncertainty-based interpretation and ranking are introduced in **Question 2**.

1.c Data Analysis and Results

To evaluate whether the hybrid Bayesian/non-Bayesian multimodal architecture could achieve the target **75–80%** classification accuracy, we conducted a **20-configuration randomized hyperparameter sweep** using **3-fold cross-validation**. Across all configurations, validation accuracy ranged from 0.63 to 0.77 (Table A4), with 11 of 20 models exceeding 0.72 and four surpassing 0.75, confirming that the target performance band is attainable under appropriate hyperparameter settings. The distribution of validation outcomes showed no pathological variance, indicating that the architecture is broadly stable under randomized hyperparameter perturbations.

Classical Performance Metrics

To illustrate contrastive behavior, we examined the highest-accuracy configuration (cfg_14) and the lowest-performing configuration (cfg_10) as representative extremal cases (Table A4). ROC curves for cfg_14 showed strong separability across nearly all genres ($\text{AUCs} \geq 0.93$), whereas cfg_10 exhibited flattened curves and reduced discriminative power, particularly for overlapping genres. Confusion matrices reinforce this distinction: cfg_14 produced a sharp diagonal with limited genre leakage, while cfg_10 showed substantial off-diagonal spillover. Training curves further differentiated the two cases—cfg_14 demonstrated smooth, stable convergence, while cfg_10 exhibited shallow learning and early plateauing. **Figure 4** and **Figure 5** are included as examples of the accuracy metrics produced by the model – a full discussion of each is included in **Appendix A**.

Comparison of High- and Low-Performing Configurations

To illustrate contrastive behavior, we examined the **best-scoring configuration (cfg_14)** and the **lowest-scoring configuration (cfg_10)** as representative extremal cases. ROC curves for cfg_14 showed strong separability across nearly all genres ($\text{AUCs} \geq 0.93$), whereas cfg_10 produced noticeably flatter curves and reduced discriminative power, especially in genres with overlapping spectral characteristics. Confusion matrices reinforced this distinction: cfg_14 yielded a sharper diagonal with limited genre leakage, while cfg_10 exhibited broader off-diagonal spillover and inconsistent per-genre predictions. Training trajectories also differed: cfg_14 displayed stable monotonic convergence, while cfg_10 showed shallow learning curves and early plateauing.

Bayesian Feature Diagnostics

Bayesian posterior analyses were performed on each configuration’s latent classification head, yielding posterior weight distributions and uncertainty estimates for all 20 runs. Although latent dimensions are not directly comparable across configurations, aggregated posterior statistics provide insight into global model behavior. **Table A3** summarizes, for each latent feature, the range and dispersion of absolute weight magnitudes, mean posterior variance (epistemic uncertainty), and frequency of prominence across runs. Sorting by posterior variance reveals a subset of epistemically stable features that consistently emerge in high-performing configurations and align with spectrally distinctive genres such as classical, metal, and jazz. In contrast, high-variance features appear more frequently in weaker configurations, signaling unstable or non-generalizable latent structure. These findings demonstrate that Bayesian heads offer diagnostic value beyond accuracy, motivating their formal integration into reliability-aware evaluation in Question 2. **Figure 2** and **Figure 3** are included as examples of Bayesian metrics produced by the model. Their full discussion is included in **Appendix A**.

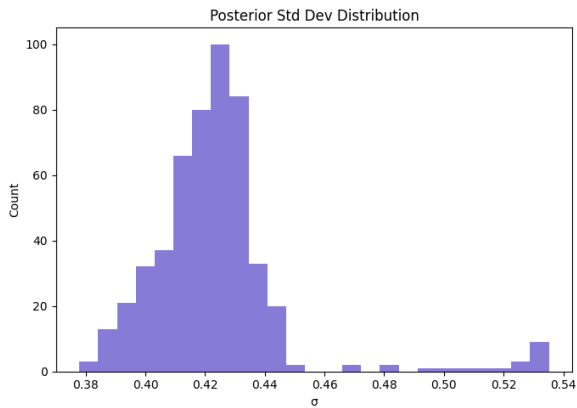


Figure 2 — Posterior σ Distribution, sweep 1, cfg_14

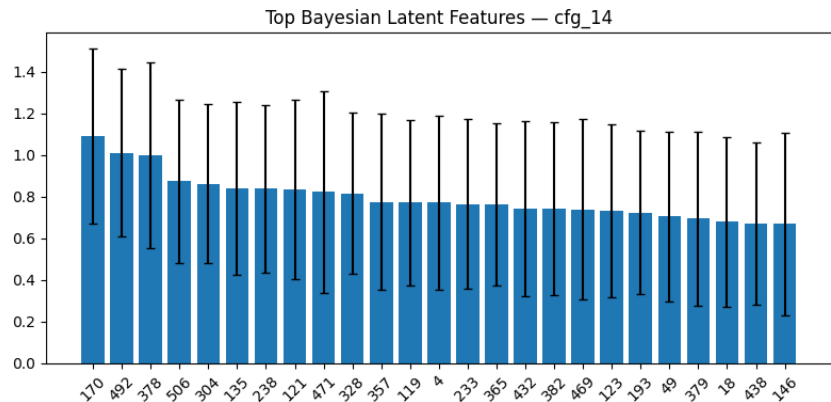


Figure 3 — Top Bayesian Latent Features, cfg_14.

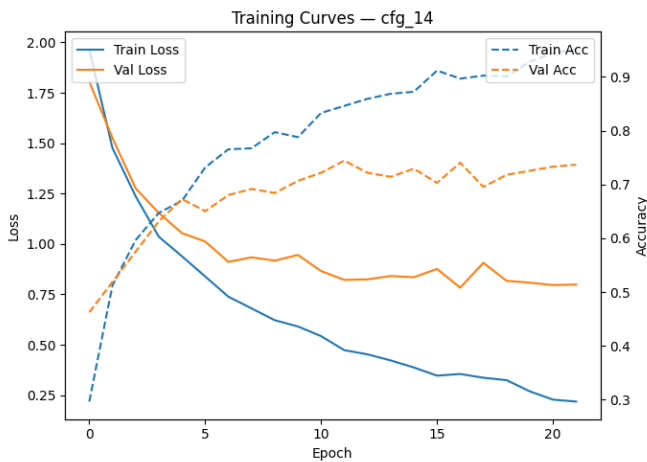


Figure 4. — Training Curves, sweep 1, cfg_14.

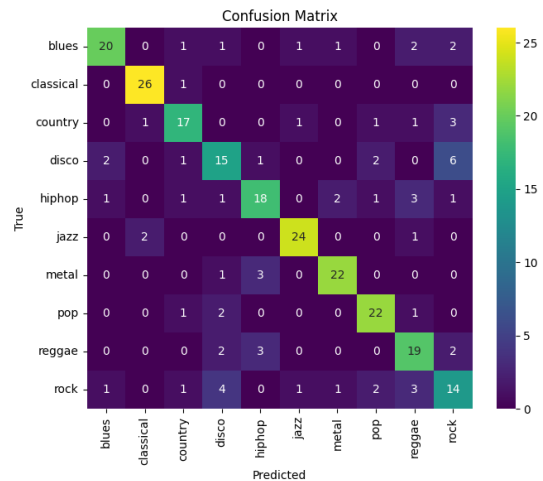


Figure 5. — Confusion Matrix, sweep 1, cfg_14.

Note:

- (1) full figure discussion, elucidation tables, and metric tables appear in appendix: Appendix – A – Question 1 – Sweep 1
- (2) Code for sweep 1 is provided in the git hub repository

1.d Discussion

Sweep-1 demonstrates that model quality depends on the interaction between validation accuracy, training-curve behavior, and Bayesian posterior variance—not accuracy alone. The models with the most promising metric configurations (e.g., `cfg_13`, `cfg_1`, `cfg_7`, and `cfg_8`) combine smooth, stable learning curves with narrow posterior σ distributions, indicating consistent decision boundaries and low epistemic uncertainty. These models are both accurate and confident for structurally meaningful reasons. We’ve learned that traditional k-fold accuracy alone is not enough for genre prediction with this architecture.

Several high-accuracy configurations - the traditional success metric - including `cfg_14`, `cfg_2`, `cfg_5`, and `cfg_15`—exhibit moderate-to-wide posterior σ , revealing latent epistemic instability despite strong validation performance (Table A5). The `cfg_14` case illustrates this clearly: although training curves show smooth convergence and high accuracy, a widening train–validation gap suggests mild overfitting. Its posterior σ distribution is centered around $\sigma \approx 0.42$ with a long tail, indicating uncertain weight estimates. The Bayesian latent-feature plot reinforces this pattern, as even the strongest latent dimensions retain large posterior intervals. While `cfg_14`’s confusion matrix shows strong overall performance, systematic blues/country/rock and disco/hip-hop confusions likely contribute to this residual uncertainty. Together, these diagnostics indicate that `cfg_14` is highly accurate but not fully reliable.

The sweep also reveals the opposite regime. Configurations such as `cfg_0` and `cfg_17` achieve relatively low accuracy yet maintain narrow posterior σ , suggesting consistent but underpowered decision boundaries. In these cases, performance limitations are architectural rather than epistemic, implying that increased model capacity—not additional regularization—would be the appropriate remedy.

Finally, configurations such as `cfg_10`, `cfg_12`, and `cfg_16` combine unstable training dynamics with very wide posterior σ . These models fail to learn coherent latent structure and represent the weakest outcomes of the sweep.

Overall, Sweep-1 shows that optimal models jointly balance accuracy and epistemic stability. In many cases, a slightly lower-accuracy model with narrow uncertainty (e.g., `cfg_7` or `cfg_8`) is more trustworthy than a higher-accuracy but epistemically diffuse model (e.g., `cfg_14`). These results demonstrate that Bayesian diagnostics provide essential insight into model confidence, reliability, and internal coherence—dimensions that are invisible to accuracy alone and motivate the reliability-aware evaluation framework developed in Question 2.

Question – 2 – Q1 extended and use of an epistemic metric:

2.a – Introduction

Question 1 showed that several models achieved strong validation accuracy using GTZAN spectrograms and audio features, yet real-world tests (e.g., *The Killers*, *Taylor Swift*) revealed severe reliability failures, often collapsing predictions into a single genre for all songs tested. These results made it clear that accuracy alone was not capturing the underlying behavior of the Bayesian neural network. Question 2 therefore has two goals: **(1)** explore whether a narrower, evidence-driven hyperparameter sweep can modestly improve accuracy, and—far more importantly — **(2)** develop evaluation metrics that integrate both frequentist performance and Bayesian epistemic uncertainty. The same dataset as Question 1 is used, but here the focus shifts from raw accuracy to model behavior across folds and posterior variability.

This shift is motivated by the fact that Bayesian neural networks inherently encode epistemic uncertainty through posterior variance [1], [4], and cross-validation research shows that mean accuracy is incomplete without considering fold-to-fold stability [2], [3]. To operationalize these principles, Question 2 introduces the Robust Ranking Metric (RRM), which combines accuracy, cross-validation stability, and posterior dispersion into a single reliability-aware score. The development of RRM is the central contribution of this section: diagnostic analyses show that RRM correlates only moderately with accuracy ($r \approx 0.62$) and very weakly with off-diagonal confusion-matrix noise ($r \approx 0.26$), demonstrating that it captures dimensions of model reliability

not reflected in accuracy. While the refined hyperparameter sweep provides comparative data, the primary goal of Question 2 is to establish metrics that better identify models that are not only accurate but also stable and epistemically trustworthy. A further exploration of more metrics that take into account both Bayesian and Frequentist properties generated by the model is considered in **Appendix - B - Question 2 - Sweep 1**, culminating in the figure: **Figure B1. Heat map of validation metrics**.

2.b Methods

To address Question 2, we apply the same multimodal architecture, preprocessing pipeline, and 3-fold cross-validation framework described in the Main Methods and used in Question 1. Models were again trained using a randomized hyperparameter sweep ($n = 20$); however, in contrast to the broad exploratory sweep of Question 1, the hyperparameter space was deliberately modified in an attempt to refine and boost accuracy scores - based on empirical performance patterns observed previously. This targeted sweep narrowed both frequentist and Bayesian parameter ranges – including **priors used** - (Appendix B, Table B2) to evaluate whether modest accuracy improvements ($\approx 2\text{--}3\%$) could be achieved while simultaneously improving fold-to-fold stability and uncertainty behavior, rather than optimizing accuracy in isolation.

In addition to the standard performance metrics computed in Question 1, Question 2 introduces and evaluates several **reliability-aware metrics** that explicitly leverage the Bayesian components of the model. These metrics are designed to give model performance feedback in the fields of: interval credibility, predictive performance, and model validation.

Chief among these is the **Robust Ranking Metric (RRM)**, which integrates mean accuracy, fold-to-fold variability, and Bayesian predictive dispersion into a single score. To contextualize RRM, two additional weight-free global comparison strategies—**Rank Aggregation** and **Pareto Front ranking**—were computed using accuracy, macro/micro performance, and Bayesian reliability measures (definitions provided in **Appendix B**). Together, these metrics enable comparison of models along multiple axes of quality, allowing Question 2 to assess not only whether accuracy improved under tighter hyperparameters, but whether resulting models were more **stable, interpretable, and epistemically well-behaved**.

2.c Data Analysis and Results

Effect of Refined Hyperparameters (Sweep 2)

Table 1 compares validation accuracy statistics between Sweep 1 and Sweep 2. Contrary to the original intent of improving performance through a narrowed, evidence-driven hyperparameter space, Sweep 2 resulted in **lower mean accuracy (0.668 vs. 0.719)** and **substantially higher variance** ($\sigma \approx 0.135$ vs. 0.035). While the maximum accuracy in Sweep 2 (0.756) remained comparable to Sweep 1, the distribution of outcomes deteriorated markedly, including several near-failure configurations with accuracy approaching chance. This outcome indicates that the refined hyperparameter constraints were **overly restrictive**, reducing the model’s ability to adapt across folds. In particular, tighter Bayesian prior ranges and regularization parameters appear to have suppressed model capacity unevenly across configurations, increasing sensitivity to initialization and fold composition. Rather than stabilizing training, the refined sweep amplified instability, demonstrating that **hyperparameter narrowing without reliability-aware feedback can degrade overall model quality**.

Accuracy Summary for Sweep 1 vs Sweep 2

Sweep	N Configs	Max Accuracy	Min Accuracy	Mean Accuracy	Std. Dev.
Sweep 1	20	0.7659	0.6283	0.7185	~ 0.035
Sweep 2	20	0.7559	0.1352	0.6684	~ 0.135

Table 1. — Accuracy comparison for sweep 1 and sweep 2 hyperparameters.

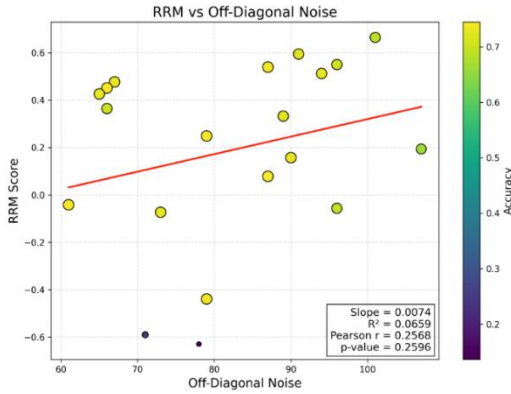


Figure 6. — RMM score vs. Off-Diagonal noise.

	Acc	σ	Umean	RRM
Acc	1.000	0.039	-0.367	0.620
σ	0.039	1.000	-0.562	-0.330
Umean	-0.367	-0.562	1.000	-0.373
RRM	0.620	-0.330	-0.373	1.000

Figure 7. — Pearson correlation Scores for Bayesian and Frequentist model metrics.

Note:

- (1) full figure discussion, elucidation tables, and metric tables appear in appendix: Appendix – B – Question 2 – Sweep 1
- (2) Code for sweep 2 is provided in the git hub repository

Accuracy vs. Reliability Trade-offs

Figure 6 (RRM vs. off-diagonal noise) and **Figure 7** (metric correlation matrix) (see Figure discussion in Appendix B) reveal why Sweep 2 performance degradation is not fully captured by accuracy alone. Accuracy shows only a **moderate correlation with RRM ($r \approx 0.62$)** and a **weak relationship with off-diagonal confusion noise ($r \approx 0.26$)**. This decoupling confirms that high accuracy does not guarantee stable decision boundaries or consistent genre separation.

Several Sweep 2 configurations achieved respectable accuracy but exhibited elevated off-diagonal noise and diffuse posterior uncertainty, leading to low RRM scores. Conversely, some lower-accuracy models maintained coherent uncertainty profiles and stable fold behavior, ranking higher under RRM. These results demonstrate that **accuracy alone obscures failure modes that become visible when uncertainty and stability are jointly considered**

Validation of the Robust Ranking Metric (RRM)

RRM proved effective at identifying models that balance performance and epistemic reliability. Configurations with high RRM scores consistently showed:

- tighter posterior σ distributions,
- lower off-diagonal confusion,
- and smoother, more repeatable training dynamics across folds.

Importantly, RRM did not simply re-rank models by accuracy: the weak correlations observed in Figure 7 confirm that RRM captures an orthogonal dimension of model quality. Models favored by RRM also aligned more closely with qualitative diagnostics such as cleaner confusion matrices and reduced genre boundary ambiguity, validating RRM as a meaningful composite metric rather than an arbitrary reweighting.

2.d Discussion

Sweep 2 demonstrates that naively refining hyperparameters based on accuracy alone can degrade both performance and stability. However, this failure directly motivates and validates the introduction of the Robust Ranking Metric (RRM). By integrating accuracy, fold-to-fold stability, and Bayesian predictive uncertainty, RRM reveals reliability differences that are invisible to conventional metrics and provides a principled basis for

model selection when accuracy alone is insufficient. Meaning – future hyperparameter sweeps and sweep tuning should involve more optimization parameters, such as the ones developed in this project.

Additional reliability-aware metrics are explored in the Appendix, including Rank Aggregation and Pareto Front Ranking (Tables B1–B2). A unified heat map visualization was also developed to evaluate model performance across multiple metrics simultaneously, enabling rapid identification of configurations that perform consistently well—or poorly—across competing objectives (Figure B1).

In summary, this study demonstrates that a CNN augmented with Bayesian inference exhibits predictive and diagnostic capabilities unavailable to purely deterministic models. Model quality in Bayesian multimodal classifiers depends not only on accuracy, but on the joint behavior of training stability, uncertainty structure, and generalization reliability—dimensions captured by RRM but obscured by accuracy alone. Future work will extend this pipeline by integrating uncertainty-aware metrics directly into adaptive hyperparameter optimization, exploring richer prior families, and optimizing explicitly for reliability-weighted performance.

References

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- [5] A. Sridhar, *Attention-Guided Spectrogram Sequence Modeling with CNNs for Music Genre Classification*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.14474>

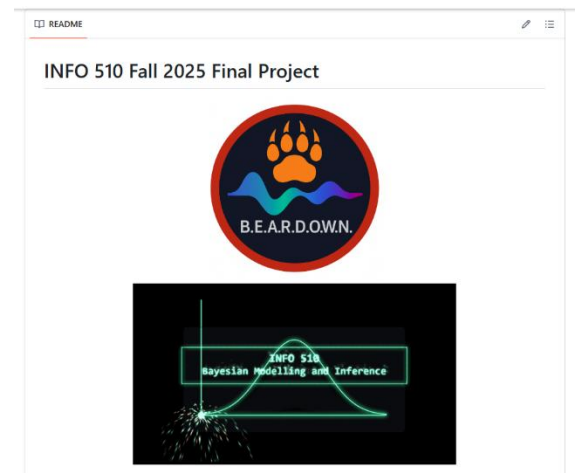
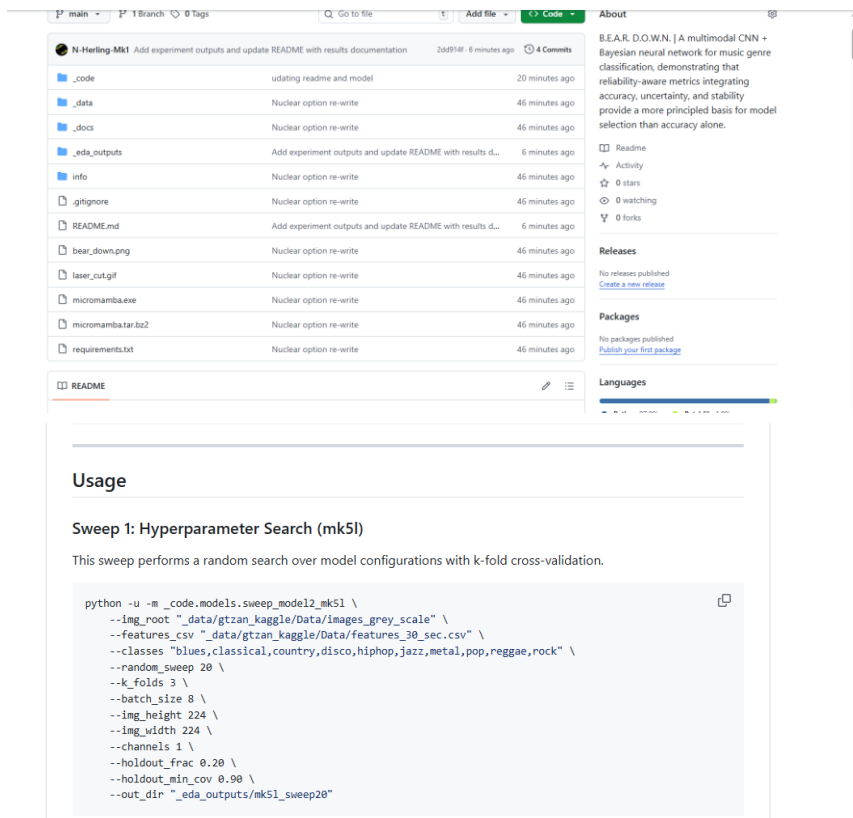
Git Hub Repo [public]
(hyper link)

[N-Herling-Mk1/INFO_510_FA_25_Final_Proj](https://github.com/N-Herling-Mk1/INFO_510_FA_25_Final_Proj)

(url)

https://github.com/N-Herling-Mk1/INFO_510_FA_25_Final_Proj

(screen shots)



Appendix – A – Question 1 – Sweep 1

Name	Range / Choices	Frequentist / Bayesian
dropout_backbone	0.15 $\rightarrow$ 0.35	Frequentist
dropout_tab	0.10 $\rightarrow$ 0.40	Frequentist
dense_units_img	{256, 384, 512, 768}	Frequentist
dense_units_tab	{64, 128, 192, 256}	Frequentist
lr (learning rate)	$10^{-4} \rightarrow 10^{-3}$	Frequentist
epochs	{50, 75}	Frequentist
optimizer	{"adam", "adamw", "rmsprop"}	Frequentist
fusion	{"concat", "gated"}	Frequentist
head_type	{"flipout", "mc_dropout", "det"}	Hybrid
kl_factor	0.9 $\rightarrow$ 1.5	Bayesian
prior_scale	0.7 $\rightarrow$ 1.5	Bayesian
posterior_scale_init	0.005 $\rightarrow$ 0.025	Bayesian

Table A1. Hyperparameters, their ranges, and types for sweep 1.

Figure discussion

(Fig. 2 — Posterior σ Distribution – Sweep 1):

The posterior standard-deviation histogram for `cfg_14` reveals a **moderate-to-wide epistemic spread** centered around $\sigma \approx 0.42$, with a noticeable heavy tail. This indicates that although the model performs well, many latent-feature weights remain only weakly constrained, reflecting **internal uncertainty** despite strong validation accuracy.

(Fig. 3 — Top Bayesian Latent Features – Sweep 1):

The top-feature bar chart shows **large posterior intervals** across `cfg_14`'s highest-magnitude latent features. This confirms that even the most influential components have **substantial epistemic variance**, consistent with the broader σ distribution and hinting that the model's internal representation is somewhat unstable across weight samples.

(Fig. 4 — Training Curves – Sweep 1):

The training curves demonstrate **smooth convergence** in both loss channels and steadily improving validation accuracy, yet the widening train–validation accuracy gap suggests **incipient overfitting**. Combined with the Bayesian diagnostics, this reinforces that the model generalizes well but is not fully confident in how it arrives at those decisions.

(Fig. 5 — Confusion Matrix – Sweep 1):

The confusion matrix shows that `cfg_14` performs **consistently across most genres**, with strong diagonals for classical, jazz, metal, and reggae. However, persistent misclassification among blues/country/rock and disco/hiphop indicates **genre-boundary ambiguity**, which may contribute to the moderate posterior uncertainty observed above.

Appendix – A – Question 1 – Sweep 1

Genre	Precision Mean	Precision 95% CI	Recall Mean	Recall 95% CI	F1 Mean	F1 95% CI	Accuracy Mean	Accuracy 95% CI
blues	0.771	[0.724, 0.819]	0.731	[0.653, 0.809]	0.728	[0.681, 0.776]	0.731	[0.653, 0.809]
classical	0.900	[0.866, 0.934]	0.901	[0.854, 0.947]	0.893	[0.871, 0.916]	0.901	[0.854, 0.947]
country	0.674	[0.621, 0.727]	0.647	[0.584, 0.710]	0.643	[0.602, 0.684]	0.647	[0.584, 0.710]
disco	0.642	[0.577, 0.707]	0.653	[0.562, 0.744]	0.616	[0.550, 0.682]	0.653	[0.562, 0.744]
hiphop	0.783	[0.729, 0.837]	0.592	[0.546, 0.638]	0.661	[0.638, 0.684]	0.592	[0.546, 0.638]
jazz	0.862	[0.807, 0.916]	0.848	[0.773, 0.923]	0.834	[0.794, 0.874]	0.848	[0.773, 0.923]
metal	0.792	[0.746, 0.837]	0.832	[0.792, 0.872]	0.806	[0.774, 0.838]	0.832	[0.792, 0.872]
pop	0.774	[0.738, 0.810]	0.775	[0.741, 0.810]	0.772	[0.744, 0.799]	0.775	[0.741, 0.810]
reggae	0.677	[0.639, 0.715]	0.654	[0.605, 0.703]	0.654	[0.631, 0.678]	0.654	[0.605, 0.703]
rock	0.505	[0.437, 0.574]	0.474	[0.398, 0.549]	0.460	[0.413, 0.508]	0.474	[0.398, 0.549]

Table A2. Per class averages, interval credibility, for sweep 1.

Bayesian Latent Feature Summary Across 20 Runs (Sorted by Mean Posterior Variance)

Feature ID	Min w	Max w	Mean w	Std w	Mean Posterior Variance	Std Posterior Variance	Count (Appeared in Runs)
1	cfg_2	0.008975	0.015350	0.012371	0.001991	0.006375	0.161
2	cfg_8	0.007843	0.016868	0.012961	0.002086	0.009025	0.161
3	cfg_9	0.007843	0.016868	0.012961	0.002086	0.009025	0.161
4	cfg_20	0.011520	0.021738	0.016783	0.002379	0.010218	0.142
5	cfg_17	0.007609	0.023895	0.017268	0.003728	0.016286	0.216
6	cfg_11	0.013474	0.023456	0.018327	0.002743	0.009982	0.150
7	cfg_16	0.014142	0.025037	0.018888	0.002487	0.010895	0.132
8	cfg_13	0.016172	0.025221	0.020685	0.002583	0.009049	0.125
9	cfg_0	0.010825	0.025688	0.020874	0.003790	0.014863	0.181
10	cfg_6	0.019620	0.029911	0.024281	0.002649	0.010291	0.109
11	cfg_18	0.020662	0.028997	0.024404	0.002213	0.008335	0.091
12	cfg_15	0.022410	0.048114	0.032127	0.006552	0.025704	0.204
13	cfg_10	0.027614	0.049009	0.038183	0.006056	0.021395	0.159
14	cfg_12	0.034058	0.049543	0.040673	0.003283	0.015485	0.081
15	cfg_1	0.054414	0.069010	0.062548	0.004240	0.014596	0.068
16	cfg_3	0.086376	0.107849	0.099009	0.005148	0.021473	0.052
17	cfg_14	0.145176	0.233713	0.171104	0.018745	0.088537	0.110
18	cfg_4	0.152281	0.186971	0.171712	0.008693	0.034690	0.051
19	cg_7	0.149503	0.237443	0.177207	0.021368	0.087940	0.121
20	cfg_5	0.192091	0.316039	0.234971	0.026881	0.123948	0.114

Table A3. Bayesian Latent feature summary.

Table 3 (Sweep 1). This table ranks latent Bayesian features by their mean posterior variance; a measure of epistemic stability that reflects how confidently the model estimates each latent dimension across runs. The $|w|$ values represent the absolute magnitude of each feature’s learned weight—its contribution strength—while sorting by posterior variance highlights which latent components are both influential and reliably inferred by the Bayesian classifier. Although latent-feature indices do not align across models, features that consistently exhibit similar posterior-variance signatures across runs likely represent functionally similar components of the learned

audio representation. These variance patterns allow us to infer stability, ambiguity, and the relative importance of different latent structures even without one-to-one feature correspondence.

Appendix – A – Question 1 – Sweep 1

Validation Accuracy Rankings		
Rank	Config	Val Acc
1	cfg_14	0.7659
2	cfg_4	0.7609
3	cfg_13	0.7597
4	cfg_1	0.7484
5	cfg_3	0.7422
6	cfg_19	0.7397
7	cfg_8	0.7384
8	cfg_11	0.7371
9	cfg_7	0.7359
10	cfg_9	0.7233
11	cfg_2	0.7196
12	cfg_5	0.7196
13	cfg_15	0.7109
14	cfg_16	0.6972
15	cfg_17	0.6908
16	cfg_6	0.6908
17	cfg_18	0.6796
18	cfg_0	0.6745
19	cfg_12	0.6371
20	cfg_10	0.6283

Table A4. Accuracy results from sweep 1.

The high-accuracy model (**Fig. A1**) shows uniformly strong ROC curves with $\text{AUC} \approx 0.94\text{--}1.00$, indicating clean class separation and low false-positive rates across all genres. Even harder classes like reggae and rock remain above 0.90, reinforcing that `cfg_14`'s boundaries are confident and well-calibrated. The low-accuracy model (**Fig. A2**) displays much weaker ROC behavior, with AUCs dropping into the **0.70–0.85** range and some classes nearing chance. Its shallow curves and early false-positive rise reveal broad confusion between genres, consistent with its unstable training dynamics and wide Bayesian σ . Overall, ROC evidence supports the sweep finding: stable, high-accuracy models form clear discriminative boundaries, while low-accuracy, high-uncertainty models do not.

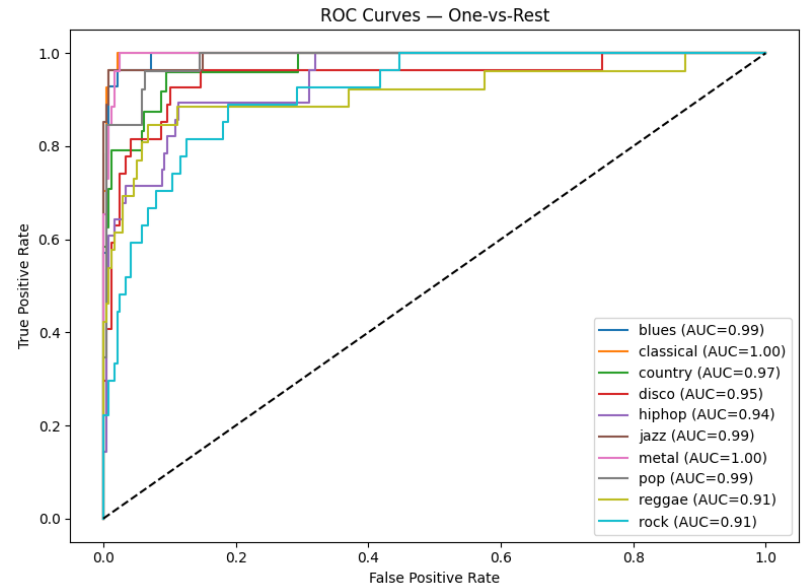


Fig. A1 – ROC Curve best accuracy sweep1 [cfg_14]

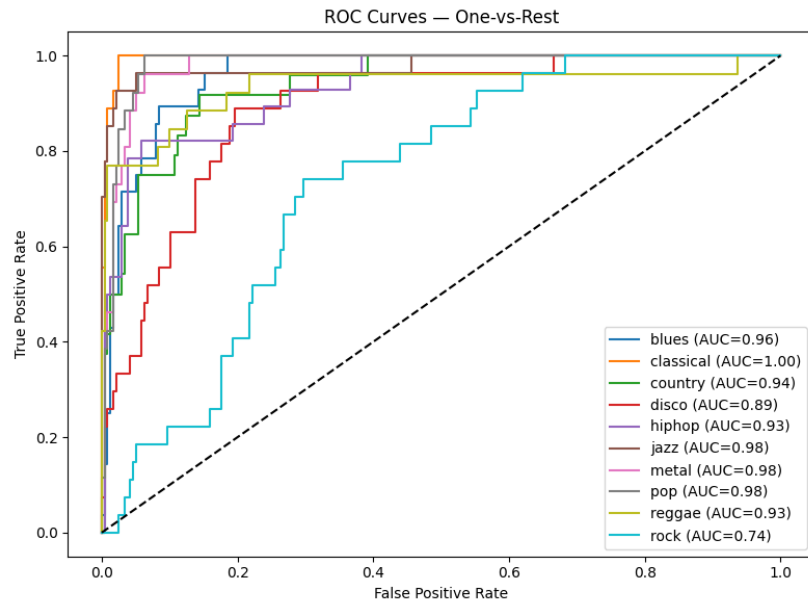


Fig. A2 – ROC Curve worst accuracy sweep1 [cfg_10]

Appendix – A – Question 1 – Sweep 1

Rank	Config	Val Acc	Training Curve Pattern	Bayesian σ Pattern	Interpretation / Notes
1	cfg_14	0.7659	Stable-ish, moderate oscillations	σ moderate-wide (0.14–0.18)	High accuracy but elevated σ → mild mismatch
2	cfg_4	0.7609	Good curves, modest gap	σ moderate (0.13–0.17)	Accuracy + σ both solid → agreement
3	cfg_13	0.7597	Very stable validation	σ narrow (0.10–0.15)	Strong agreement: high accuracy + low σ
4	cfg_1	0.7484	Smooth curves	σ narrow (0.10–0.13)	Clean signal: reliable generalization
5	cfg_3	0.7422	Stable, low noise	σ narrow (0.11–0.16)	Very good alignment
6	cfg_19	0.7397	Moderate oscillations	σ moderate (0.13–0.20)	Good accuracy but σ rising → slight mismatch
7	cfg_8	0.7384	Strong stability	σ narrow (0.14–0.16)	Very clean Bayesian profile
8	cfg_11	0.7371	Smooth validation	σ narrow–medium (0.13–0.18)	Good balance
9	cfg_7	0.7359	Stable curves	σ narrow (0.11–0.14)	Agreement: trustworthy model
10	cfg_9	0.7233	Some spikes but OK	σ moderate (0.13–0.17)	Accuracy fair, σ acceptable
11	cfg_2	0.7196	Higher oscillations, larger gap	σ wide (0.18–0.26)	High-ish accuracy but uncertainty too high
12	cfg_5	0.7196	Chaotic val curve spikes	σ wide (0.20–0.30)	Mismatch: accuracy high but poor uncertainty
13	cfg_15	0.7109	Large instabilities	σ wide (0.45–0.60)	One of the worst σ profiles → conflict
14	cfg_16	0.6972	Noisy, divergent curves	σ very wide (0.30–0.50+)	Low accuracy + high σ → bad model
15	cfg_17	0.6908	Stable-ish but lower accuracy	σ narrow (0.13–0.17)	Low accuracy but surprisingly good σ
16	cfg_6	0.6908	Extremely unstable curves	σ very wide (0.30–0.40+)	Poor accuracy + poor σ → discard
17	cfg_18	0.6796	Smooth-ish but underfits	σ moderate-wide (0.15–0.20)	Accuracy low but uncertainty reasonable
18	cfg_0	0.6745	Stable, consistent	σ narrow (0.10–0.13)	Low accuracy but excellent calibration
19	cfg_12	0.6371	Very unstable	σ wide (0.20–0.35+)	Weak accuracy + weak Bayesian stats
20	cfg_10	0.6283	Unstable & divergent	σ very wide (0.22–0.27)	Worst combination

Table A5 -sweep 1. Total summary of epistemic variance, training curve analysis.

Across the 20-run sweep, the highest-performing models (e.g., *cfg_14*, *cfg_4*, *cfg_13*) generally show strong agreement between validation accuracy and Bayesian uncertainty—stable training curves paired with narrow or moderately narrow posterior σ distributions. Several mid-ranked models (such as *cfg_2*, *cfg_5*, *cfg_15*) achieved respectable accuracies but exhibited unusually wide σ ranges, signaling that their predictions are far less reliable despite decent performance. At the lower end, configurations like *cfg_10*, *cfg_12*, and *cfg_6* showed both poor accuracy and large posterior variance, making them clear candidates for exclusion in future tuning cycles.

Low posterior variance corresponds to lower epistemic uncertainty, indicating that the Bayesian classifier assigns stable, consistent predictions across MC samples. Models with narrow σ -distributions tend to represent genres more cleanly in latent space and display more reliable decision boundaries, even in cases where accuracy is not maximal. In contrast, wide posterior σ values reflect unstable or ambiguous representations, often accompanying higher off-diagonal confusion and weaker generalization.

Appendix – B – Question 2 – Sweep 1

Model Penalty Vector

$$v = \left[1 - A, \frac{s}{s_{\max}}, \frac{U}{U_{\max}} \right]$$

Penalty Norm

$$\|v\| = \sqrt{(1 - A)^2 + \left(\frac{s}{s_{\max}}\right)^2 + \left(\frac{U}{U_{\max}}\right)^2}$$

Final Metric (Reward Form)

$$\text{RRM} = 1 - \|v\| \quad \text{--- (1)}$$

Metric Range (Numerical)

$$\begin{aligned} \|v\|_{\min} = 0 &\Rightarrow \text{RRM}_{\max} = 1 \\ \|v\|_{\max} = \sqrt{3} \approx 1.732 &\Rightarrow \text{RRM}_{\min} = 1 - \sqrt{3} \approx -0.732 \end{aligned}$$

Figure B1. RRM definition

Symbol	Meaning	Type
A	Accuracy of the model (higher = better)	Frequentist
1 - A	Penalty: distance from perfect accuracy	Frequentist
s	Cross-fold standard deviation of validation accuracy	Frequentist
s _{max}	Worst (largest) instability across models	Frequentist
U	Epistemic uncertainty (posterior spread)	Bayesian
U _{max}	Worst epistemic uncertainty observed	Bayesian
v	Combined normalized penalty	Frequentist × Bayesian
RRM	Reward score: 1 - v	Hybrid

Table B1. RRM salients

Name	Range / Choices (Sweep 2)	Frequentist / Bayesian	Change from Sweep 1
dropout_backbone	0.10 → 0.28	Frequentist	Narrowed from (0.15–0.35) to tighter high-performance interval.
dropout_tab	0.10 → 0.25	Frequentist	Upper bound reduced from 0.40 due to over-regularization.
dense_units_img	{256, 384, 512}	Frequentist	Removed 768 — diminishing returns in Sweep 1.
dense_units_tab	{128, 192, 256}	Frequentist	Removed 64-unit option — persistent underperformer.
lr	10 ⁻⁴ → 10 ^{-3.35}	Frequentist	Constrained to best-performing region.
epochs	{50, 75, 100, 125, 150}	Frequentist	Expanded upward to exploit early stopping gains.
optimizer	{"adam", "adamw"}	Frequentist	Removed RMSprop; weighted toward Adam.
fusion	{"gated", "concat"}	Frequentist	Weighted strongly toward gated fusion.
head_type	{"det", "mc_dropout", "flipout"}	Hybrid	Reduced Flipout usage; deterministic preferred.
kl_factor	1.0 → 1.5	Bayesian	Slight upward shift vs Sweep 1.
prior_scale	0.7 → 1.5	Bayesian	Unchanged from Sweep 1.
posterior_scale_init	0.005 → 0.03	Bayesian	Slightly widened upper bound (0.025 → 0.03).

Table B2. (Assumed) Optimized Hyperparameter Space After Sweep 1 Insights.

Figure discussion

(Table 1. — Accuracy comparison for sweep 1 and sweep 2 hyperparameters.)

Table 1 compares validation accuracy statistics for the broad (Sweep 1) and refined (Sweep 2) hyperparameter searches. While Sweep 2 achieves a comparable maximum accuracy, its mean accuracy decreases substantially and variance increases, indicating reduced overall stability across configurations. This contrast highlights that narrowing hyperparameter ranges without uncertainty-aware guidance can negatively impact both average performance and robustness.

Appendix – B – Question 2 – Sweep 1

(Fig. 6 — RRM score vs. Off-Diagonal noise.)

Figure 6 — RRM score vs. Off-Diagonal noise.

This figure shows the relationship between the Robust Ranking Metric (RRM) and off-diagonal noise across all configurations. The weak positive correlation ($r \approx 0.26$) indicates that RRM does not simply track confusion-matrix volatility or epistemic error. Instead, RRM captures reliability information that is largely orthogonal to off-diagonal noise, supporting its role as a complementary evaluation metric.

(Fig. 7 — Pearson correlation Scores for Bayesian and Frequentist model metrics.)

Figure 7 — Pearson correlation scores for Bayesian and frequentist model metrics.

This correlation matrix summarizes relationships among accuracy, posterior uncertainty (σ), mean predictive uncertainty (Umean), and RRM. RRM exhibits a moderate positive correlation with accuracy ($r \approx 0.62$) while remaining negatively correlated with uncertainty measures, demonstrating that it balances performance and epistemic stability rather than collapsing into a proxy for any single metric. These relationships confirm that RRM integrates both frequentist and Bayesian signals into a distinct, reliability-focused score.

Appendix – B – Question 2 – Sweep 1

Rank Aggregation Metric (RAM)

Combining independent model rankings into a unified score

$$\text{RAM}(i) = \text{rank_A}(i) + \text{rank_R}(i) + \text{rank_M}(i) + \text{rank_U}(i)$$

(2)

Component	Description
rank_A(i)	Rank of model i by validation accuracy.
rank_R(i)	Rank of model i by RRM (reliability-adjusted performance).
rank_M(i)	Rank based on macro-averaged per-genre score (treats all genres equally).
rank_U(i)	Rank based on micro-averaged metrics (weighted by sample counts).
RAM(i)	Total aggregated rank; lower values indicate superior global performance.
Min/Max Interpretation	The minimum possible RAM value occurs when a model ranks 1st in every metric (ideal). Higher RAM values indicate increasing disagreement across metrics and weaker overall performance.

Table B1. Rank Aggregation Metric

The Rank Aggregation Metric combines four independent orderings—accuracy, RRM, macro score, and micro score—into a single unified ranking by summing each model’s position in the individual lists. A model with consistently strong results will receive low ranks in all categories, causing its RAM score to approach the minimum possible value (best-case = 4). Conversely, models that excel in only one dimension and underperform in others accumulate larger rank-sums, pushing the metric upward and indicating weaker global performance. Because each component rank is normalized relative to the full cohort, RAM provides a stable, weight-free integration of metrics, rewarding models that are not only accurate but also reliable and fair across genres. Models with lower RAM values are therefore globally superior even if their raw accuracy is not the absolute highest.

Appendix – B – Question 2 – Sweep 1

Pareto Front Rank (PFR)

Ranking models by joint accuracy–reliability optimality

Model i dominates model j if:
 $A(i) \geq A(j)$ and $R(i) \geq R(j)$,
and at least one inequality is strict. (3)

PFR(i) = front index of model i (4)

Component	Description
$A(i)$	Validation accuracy.
$R(i)$	RRM score (Bayesian reliability).
Dominance(i, j)	True if model $i \geq j$ in both accuracy and reliability, and strictly better in one.
Pareto Front 1	The set of all non-dominated models (optimal frontier).
PFR(i)	Front number assigned to model i ; 1 = non-dominated, higher = increasingly dominated.
Min/Max Interpretation	The minimum possible PFR value is 1 , indicating full Pareto optimality (non-dominated across all metrics). Higher PFR values represent models that are increasingly dominated and lie farther from the optimal accuracy–reliability frontier.

Table B2. Pareto ranking metric.

Pareto Front analysis evaluates models jointly by accuracy and reliability (RRM), identifying those that are *non-dominated*—no other model is strictly better in both dimensions. Models on **Pareto Front 1** achieve the optimal balance: they cannot be improved in one metric without losing performance in the other, and thus represent the best accuracy–stability trade-offs. Higher-tier fronts (PFR = 2, 3, ...) reflect increasingly dominated configurations whose performance profiles are inferior in both accuracy and RRM to some model on a lower tier. Because the method is order-based rather than value-based, no min/max scaling is required: a model's tier directly reflects how close it lies to the theoretical optimum boundary of high accuracy and low epistemic uncertainty. PFR therefore captures the *shape* of performance trade-offs, highlighting models that are simultaneously strong classifiers and epistemically stable.

Appendix – B – Question 2 – Sweep 1

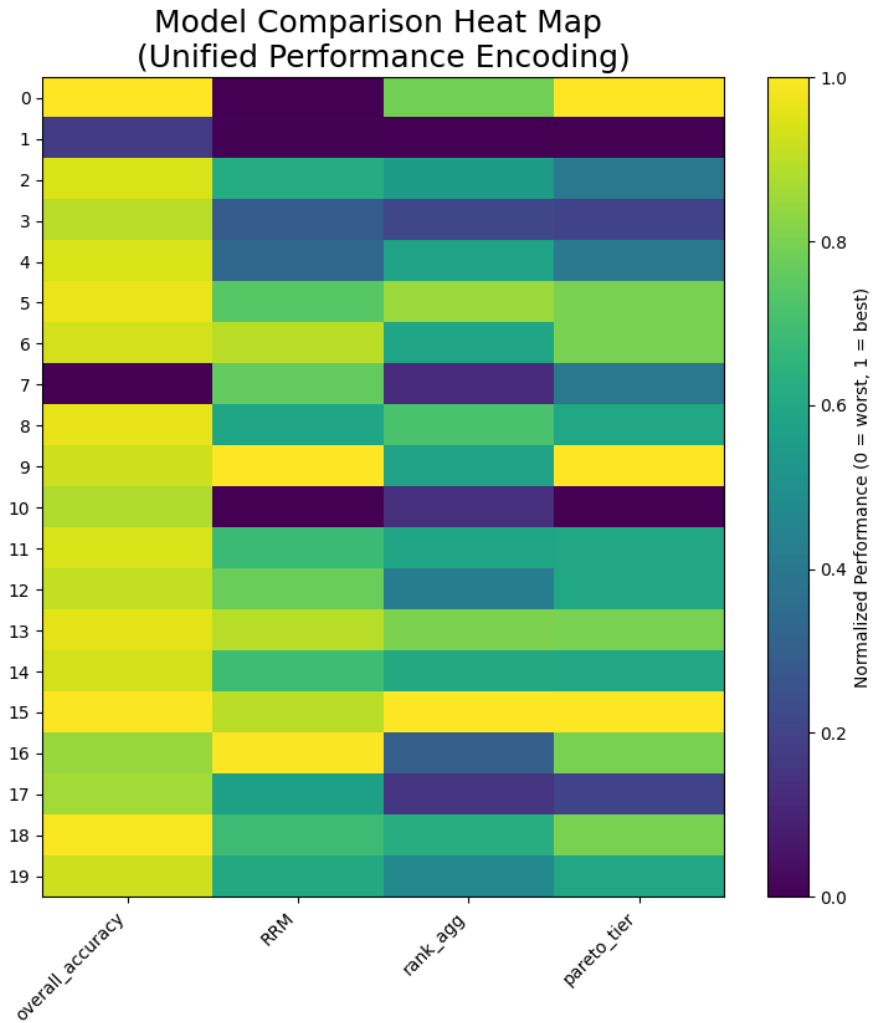


Figure B1. Heat map of validation metrics.

Across all four normalized performance dimensions—overall accuracy, RRM reliability, rank aggregation, and Pareto optimality—the heat map reveals a clear separation between the strongest and weakest model configurations. **Configurations 15, 9, 11, and 14 emerge as the top performers**, each displaying consistently bright, high-scoring cells across the full metric set, indicating balanced excellence in accuracy, reliability, and multi-objective stability. **Among these, cfg 15 stands out as the most uniformly strong.** In contrast, **cfg 7, cfg 10, and cfg 1** form the lowest tier, with dark, low-performing values across multiple metrics—particularly in accuracy and RRM—signaling underperformance both individually and in multi-metric evaluation. This visual clustering demonstrates not only which models are strongest overall but also how consistently each configuration performs across competing criteria.

Appendix – C – Model Description

Layer	Type	Description	Components Included
Layer A Dual Encoders	Deterministic	Two parallel encoders convert multimodal inputs (spectrogram + tabular features) into compact latent embeddings.	Spectrogram CNN - Conv → ReLU → Pool blocks - Flatten → Dense embedding - Dropout (deterministic) Tabular MLP - Input: 58 normalized CSV features - Dense → ReLU layers - Dropout (deterministic)
Layer B Fusion Block	Fusion / Mixed	Combines CNN and tabular embeddings into a single multimodal latent representation.	- Concatenation layer - Gated Fusion: sigmoid(gate) × tabular - Optional normalization - Produces fused vector z_{fused} - Supports multimodal blending - Stable across feature scales
Layer C Classification Head	Bayesian / Probabilistic	Produces class probabilities and epistemic/predictive uncertainty.	Head Variants - Deterministic Dense → Softmax - MC-Dropout Bayesian head - Flipout Bayesian Dense Outputs 10-class softmax - Epistemic & total uncertainty - MC accuracy distributions - Full per-genre metrics

Table 1C. Multimodal Bayesian Network Architecture and Layer Roles

This table summarizes the three-stage architecture used across both sweeps, highlighting the separation between deterministic feature extraction and Bayesian uncertainty modeling. Dual deterministic encoders first transform spectrogram images and tabular audio features into compact latent embeddings, which are then merged via a fusion block to produce a unified multimodal representation. Crucially, epistemic and predictive uncertainty are introduced only at the classification head through Bayesian layers (MC Dropout or Flipout), ensuring that uncertainty estimates reflect ambiguity in class decision boundaries rather than noise in low-level feature extraction.

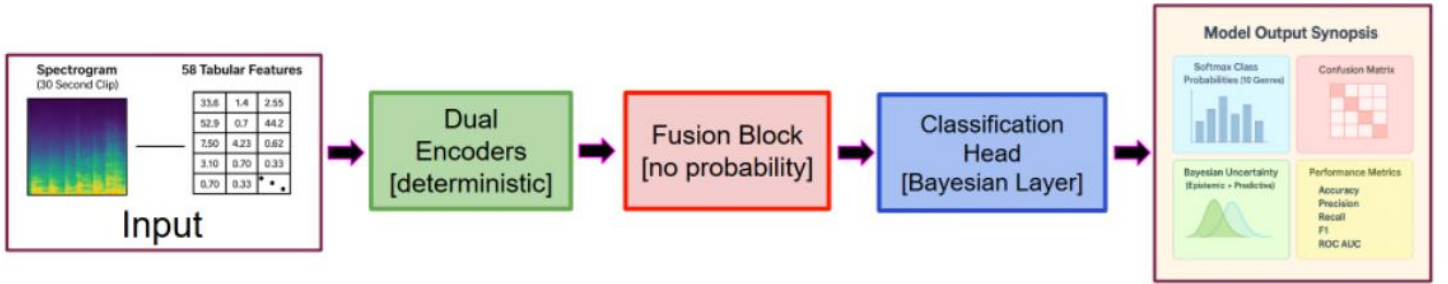


Figure 1C. Layer-Level Decomposition of the Multimodal Bayesian Classifier.

This figure details abstraction definitions of each architectural layer, explicitly distinguishing deterministic components from probabilistic ones. The encoder and fusion stages are fully deterministic, designed to learn stable, reusable latent representations from multimodal inputs, while the classification head operates in a Bayesian manner to model uncertainty over class predictions. This design choice enables principled uncertainty estimation without destabilizing early feature learning, aligning with Bayesian best practices that localize uncertainty modeling to high-level decision layers. The information flow of input spectrogram to output model along with model metrics is clearly labeled.

Note: The input spectrogram is colored in the viridis color palette for visual effect, the actual spectrograms are grey scale.

Appendix – C – Model Description

Bayesian vs. Non-Bayesian Hyperparameters Sweep 1 (mk5l) vs Sweep 2 (mk5n)

Bayesian Hyperparameters (Priors, Posterior Scale, KL)

Hyperparameter (Bayesian Only)	Sweep 1 (mk5l)	Sweep 2 (mk5n)	Description / Role
prior_scale	Uniform(0.8, 1.6)	Uniform(0.7, 1.5)	Controls variance of Gaussian weight priors — broader priors encourage higher epistemic uncertainty.
posterior_scale_init	Uniform(0.005, 0.02)	Uniform(0.005, 0.03)	Initialization range for posterior log-variance; Sweep 2 allows broader initial uncertainty.
kl_factor	Uniform(1.0, 1.6)	Uniform(1.0, 1.5)	Controls KL divergence weight in the ELBO objective; Sweep 2 slightly restricts high-KL models.
head_type	{det, mc_dropout, flipout}	{det, mc_dropout, flipout} (reduced probability of flipout)	Controls which Bayesian last layer is used: Flipout and MC-Dropout introduce Bayesian uncertainty.

Non-Bayesian Hyperparameters (Architecture, Regularization, LR, Training)

Hyperparameter	Sweep 1 (mk5l)	Sweep 2 (mk5n)	Description / Role
dropout_backbone	U(0.18, 0.32)	U(0.10, 0.28)	Backbone regularization — Sweep 2 shifts downward to reduce oversuppression.
dropout_tab	U(0.12, 0.40)	U(0.10, 0.25)	Tabular branch dropout — Sweep 2 tightens range for stability.
dense_units_img	{384, 512, 768}	{256, 384, 512}	Image dense layer width; Sweep 2 lowers top-end capacity.
dense_units_tab	{96, 128, 192, 256}	{128, 192, 256}	Tabular dense width; Sweep 2 removes low-capacity values.
Learning Rate	$10^4 U(-3.8, -3.0)$	$10^4 U(-4.0, -3.35)$	Shift toward slower rates observed to reduce epistemic noise.
Epochs	{50, 75}	{50, 75, 100, 125, 150}	Sweep 2 allows deeper training for more stable posterior convergence.
Optimizer	{adam, adamw, rmsprop}	{adam, adamw}	Sweep 2 removes RMSProp due to instability in Sweep 1.
fusion	{concat, gated}	{gated, concat} (gated favored)	Gated fusion produced more stable Bayesian posteriors, so Sweep 2 biases toward it.

Table 2C. Bayesian (priors) and Non-Bayesian Hyperparameter Design Across Sweep 1 and Sweep 2.

This table summarizes the full hyperparameter design for both sweeps, explicitly distinguishing Bayesian components—priors, posterior initialization, KL regularization, and Bayesian head selection—from standard architectural and training hyperparameters. Sweep 2 reflects an attempt at a more constrained, evidence-driven refinement of the Bayesian prior ranges and KL weighting to moderate epistemic uncertainty, while simultaneously tightening non-Bayesian regularization and capacity controls to improve stability (largely unsuccessful I might add). By separating these roles, the table clarifies which design choices directly affect uncertainty modeling versus optimization behavior, providing transparency into how the refined sweep targets reliability rather than accuracy alone.

Appendix – C – Model Description

Table 3C. Complete list of features. This table enumerates the full set of 58 audio features used by the model, derived from 30-second audio segments. The features span spectral, temporal, rhythmic, harmonic, chroma, and MFCC-based descriptors, with both mean and variance statistics included where appropriate to capture long-term structure and variability. Together, these features provide a compact but expressive representation of timbre, rhythm, and harmonic content for downstream genre classification.

count (n)	Feature Name	Category	Statistic
1	spectral_centroid_mean	Spectral	Mean
2	spectral_centroid_var	Spectral	Variance
3	spectral_bandwidth_mean	Spectral	Mean
4	spectral_bandwidth_var	Spectral	Variance
5	spectral_rolloff_mean	Spectral	Mean
6	spectral_rolloff_var	Spectral	Variance
7	zero_crossing_rate_mean	Temporal	Mean
8	zero_crossing_rate_var	Temporal	Variance
9	rms_mean	Energy	Mean
10	rms_var	Energy	Variance
11	chroma_stft_mean	Chroma	Mean
12	chroma_stft_var	Chroma	Variance
13	tempo	Rhythm	Global
14	mfcc1_mean	MFCC	Mean
15	mfcc2_mean	MFCC	Mean
16	mfcc3_mean	MFCC	Mean
17	mfcc4_mean	MFCC	Mean
18	mfcc5_mean	MFCC	Mean
19	mfcc6_mean	MFCC	Mean
20	mfcc7_mean	MFCC	Mean
21	mfcc8_mean	MFCC	Mean
22	mfcc9_mean	MFCC	Mean
23	mfcc10_mean	MFCC	Mean
24	mfcc11_mean	MFCC	Mean
25	mfcc12_mean	MFCC	Mean
26	mfcc13_mean	MFCC	Mean
27	mfcc14_mean	MFCC	Mean
28	mfcc15_mean	MFCC	Mean
29	mfcc16_mean	MFCC	Mean
30	mfcc17_mean	MFCC	Mean
31	mfcc18_mean	MFCC	Mean
32	mfcc19_mean	MFCC	Mean
33	mfcc20_mean	MFCC	Mean
34	mfcc1_var	MFCC	Variance
35	mfcc2_var	MFCC	Variance
36	mfcc3_var	MFCC	Variance
37	mfcc4_var	MFCC	Variance
38	mfcc5_var	MFCC	Variance
39	mfcc6_var	MFCC	Variance
40	mfcc7_var	MFCC	Variance
41	mfcc8_var	MFCC	Variance
42	mfcc9_var	MFCC	Variance
43	mfcc10_var	MFCC	Variance
44	mfcc11_var	MFCC	Variance
45	mfcc12_var	MFCC	Variance
46	mfcc13_var	MFCC	Variance
47	mfcc14_var	MFCC	Variance
48	mfcc15_var	MFCC	Variance
49	mfcc16_var	MFCC	Variance
50	mfcc17_var	MFCC	Variance
51	mfcc18_var	MFCC	Variance
52	mfcc19_var	MFCC	Variance
53	mfcc20_var	MFCC	Variance
54	spectral_contrast_mean	Spectral	Mean
55	spectral_contrast_var	Spectral	Variance
56	tonnetz_mean	Harmonic	Mean
57	tonnetz_var	Harmonic	Variance
58	chroma_cqt_mean	Chroma	Mean

Appendix D – ai usage statement



Generative AI Tool Use Acknowledgment:

Generative AI tools, including **OpenAI’s ChatGPT** and **Microsoft Copilot (image generation)**, were used to support this project. ChatGPT assisted with clarifying statistical concepts, refining written sections, organizing report structure, and generating explanatory text, while all analytical decisions, coding, and interpretation of results were performed independently by the author. The use of these tools followed an iterative prompting process, where multiple refinements were required to reach accurate, context-appropriate outputs; no single prompt produced a complete or final solution. Microsoft Copilot was used solely for generating illustrative images that supported conceptual understanding. All final methodological choices, analyses, and conclusions reflect the author’s own work and judgment.